

**From Reactive to Predictive: A Lifecycle Strategy for Clinical Endpoint Devices at
Stanford Health Care**

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Executive Summary

The point-of-care computing device at Stanford Health Care (SHC) is the workstation on wheels (WOW) used for barcode medication administration, clinical documentation, and specimen labeling. Throughout the health care field, the replacement schedule of these devices tends to follow an age-based cycle, although the parts of these devices do not deteriorate equally; batteries generally endure for three to five years, while the chassis are able to serve for seven to ten years (CME Corporation, 2026). Therefore, an age-based replacement strategy will retire functional hardware, while failing to replace parts that degrade in clinical operations, causing failure until the part fails, which is documented to result in a clinical outcome in the form of equipment failure being one of the key obstacles for utilization of the barcode medication administration (BCMA), and workarounds for BCMA being highly correlated with higher rates of medication administration errors (Grailey et al., 2023; van der Veen et al., 2018).

For that reason, this capstone will analyze the issue of the mismatch between the condition of the assets and the timing of their replacements as a business problem and propose the transition from age-based failure triggering asset replacement to lifecycle-based condition triggering asset management. In turn, all findings presented in this report were formulated based on the publicly available financial data of SHC, industry research papers on the service life of the clinical endpoints devices, conditions of the clinical workforce, and sustainable practices in the field of healthcare delivery. The report adheres to the conditions under which SHC allowed the use of its name for this project: no protected health information, no confidential information, no proprietary operational data, and no private metrics were included into the report without a prior written consent.

The financial justification for the suggested approach does not need the avoidance of patient harm to prove its validity. First of all, component repair for this equipment class is typically performed under vendor warranties with the schedule-driven rather than continuous maintenance contract; therefore, the most significant cost incurred in case of failure is the downtime. The proposed model of condition-based approach to maintenance simply changes the trigger to the same already existing and paid service relationship. Therefore, the condition-based approach implementation is cost-effective. Moreover, the unquantifiable but valuable advantage of such approach is the avoidance of harm; the cost per preventable adverse drug events ranges from \$3,511 to \$8,116 (Hug et al., 2012).

In turn, the impact of the issue on the clinical workforce cannot be understated either. A combined analysis of 66,177 health care professionals demonstrated that 40.4% of them experience burnout due to the use of EHR systems (Wu et al., 2024). The statistics about the labor market demonstrates that 41% of nurses who intend to stop practicing within the next five years point out stress and burnout among the top reasons second only to retirement (Whalen, 2026). The equipment failure during the medication pass is one of the contributing factors to this problem, so the question of endpoint reliability becomes a workforce issue to the equal extent as a technological one.

This capstone will cover the organizational and strategic position of SHC, the current state of the lifecycle management of the analyzed equipment class and the typical pattern of its operation, the financial benefits of the condition-based approach, its impact on the workforce and leadership of SHC, its sustainability implications and the ethics that it creates. The report concludes with the recommendation about the phased, capital-neutral implementation of the

program conditioned upon validating certain figures through SHC's data process. Such an approach will allow SHC to take action without waiting for the missing data to be provided.

Introduction

In order to demonstrate all MBA program outcomes with this capstone, the scope of the analysis should be narrow enough to be analyzed in details and wide enough to involve several business disciplines. This report meets both criteria since it involves a single asset class (the fleet of clinical endpoint devices) of a single company (Stanford Health Care) in the fields of finance, operations, workforce, sustainability, and ethics.

As a contractor, Tech Solutions Analyst for SHC, I am professionally familiar with the business environment this report explores with any organization-specific information used in it being provided under the terms of SHC's approval of the project. These terms, confirmed in writing, allow usage of SHC's name and public information but require pre-approval prior to inclusion of any protected health information, confidential information, proprietary operational data, and non-public metric. This report follows this requirement: every organizational fact is obtained from SHC's public bond disclosures, and every assertion on device management processes made here is drawn from public industry and academic literature without SHC-specific confirmation identified as a next step.

This report contains seven sections. In section I, SHC's organizational context and strategy will be defined based on SHC's public financial disclosures. In section II, the current-state endpoint lifecycle approach will be analyzed in comparison with the industry standards applicable to this equipment class. In section III, the financial case for the condition-based alternative to be introduced will be built. Section IV will focus on the implementation challenges related to SHC's personnel, its leaders and cross-organization coordination of efforts. Section V will assess the proposed change's sustainability implications. Section VI will address the ethical issues raised by the proposed change and that the change introduces. Section VII will offer phased recommendations based on the evidence presented above with stated conditions and success measures.

I. Organizational Context and Strategic Position

Stanford Health Care is a not-for-profit, academic medical center headquartered in Palo Alto, California, and is a wholly owned subsidiary of Stanford University. SHC's stated mission is "to care, to educate, and to discover," and its vision statement is "healing humanity through science and compassion, one patient at a time" (Stanford Health Care, n.d.). Besides the main campus in Palo Alto, SHC operates Stanford Health Care, Tri-Valley in Pleasanton, and a regional ambulatory network featuring Stanford faculty physicians and physicians affiliated with Stanford through Stanford Medicine Partners program.

SHC is a large organization with extraordinary clinical complexity. SHC is licensed for 622 acute-care beds with 17,777 full-time equivalent staff and 3,141 active physicians as of fiscal year 2024. During fiscal year 2024, SHC served 45,264 inpatients, recorded 155,393 emergency visits, performed 15,525 inpatient surgeries and 41,617 outpatient surgeries (S&P Global Ratings, 2025). Its Medicare case mix index of 2.53 is exceptionally high and points at tertiary and quaternary service predominance, SHC serves patient population that is much more complex than that of an ordinary community hospital.

Fiscally, SHC was able to generate \$8.95 billion of operating revenue versus \$8.53 billion of operating expenses in fiscal year 2024, earning operating income of \$424.2 million and operating margin of 4.74%. Commercial-weighted payer mix includes 61.7% commercial and Blues plans,

24.5% Medicare, and 8.9% Medicaid. According to S&P Global Ratings (2025), SHC has been assigned an AA- long-term rating with a positive outlook due to strong cash flow generation, brand recognition and advantage from the university affiliation.

The main advantage of SHC is its main limitation: demand. The average number of patients per day varied from 655 to 723, with only 622 licensed beds, meaning that the company operates under a state capacity waiver. This is crucial for the following analysis. The difference is that an organization operating above licensed capacity and treating patients twice as complicated as the average Medicare case is much less flexible in terms of absorbing equipment failure compared to a lower acuity system. A failure, which the community hospital could absorb without any consequences, falls upon SHC units providing treatment to the most complicated cases with no margin of buffer capacity.

SHC is the only competing organization in the high acuity referrals business with UCSF. In addition, the organization faces growing ambulatory competition from Sutter Health in the southern Santa Clara County. SHC has an extensive capital program: renovation of the 300 Pasteur facility with the cost of \$1.6 billion and \$711 million Redwood City outpatient expansion project will be finished before the 2030 California life safety compliance deadline (S&P Global Ratings, 2025). A capital proposal competing for capital resources at SHC now means that this proposal has to compete with a mandatory life safety construction project, which is much easier for a capital light proposal to win.

Strategic Implications for This Analysis

Each of these facts is not just a contextual element, but the actual criteria that have to be taken into account when choosing a solution. An organization that sets as its mission to care, to educate, and to discover (Stanford Health Care, n.d.) perceives equipment reliability as the part of the mission itself: failing to provide reliable equipment during medication administration means failing to care about patients. An organization competing on reputation against UCSF cannot afford failures that lower reputation profile competitors could absorb without losing the reputation. And an organization investing in two projects for the amount of \$2.3 billion has, in any case, less desire for additional capital request than usual, which is why the solution developed in Section III is capital-neutral, not intensive. Each further section of the report refers to these three organizational facts as the criteria for the evaluation of the proposed solution.

It should be also noticed that there is something this picture does not show: SHC is not a financially troubled institution looking for cost reduction opportunities; it is a highly rated, developing academic medical center with good margins in comparison with the sector's peers. This fact is important for understanding the context of the proposal. The proposal is not a rescue plan for troubled institutions, but the investment in maturity of successful one. The type of operation refinement which healthy organization is able to do because it is not experiencing acute financial difficulties to postpone it.

SHC's market position also defines the real target audience of this analysis. SHC covers Santa Clara and San Mateo counties' markets with total population of 2.66 million, and it holds a significant but not dominant position on the market in comparison with UCSF and the developing ambulatory market of Sutter Health (S&P Global Ratings, 2025). Being in the market with high competition with a choice of different options for both patients and referring physicians, the reliability failure that becomes visible for either one of these groups carries a cost of reputation that would not occur in low-competition market. This is not the main reason of the

recommendation made in this report, based on the economics of equipment described in Section III, but the secondary one in the context of the audience for whom this report is written: reputation risk management is the legitimate input for capital allocation decision, even if it cannot be quantified precisely.

II. Current-State Analysis of the Endpoint Lifecycle Model

SHC uses workstations on wheels (WOWs) as the point-of-care computing platform through which nurses complete barcode medication administration (BCMA), document care, and label specimen samples. Similar to what other academic medical centers do, the replacement policy for WOWs is based on calendar age (CME Corporation, 2026). The key analytical insight from this section is that the calendar age replacement policy is misaligned with the actual degradation patterns of this equipment, and that misalignment, not any design flaws of the devices, is what causes the reliability issue that this report aims to solve.

The Service-Life Mismatch

By design, WOWs are modular systems: the cart chassis, computing unit, battery, and scanner are all replaceable individually instead of being assembled into one sealed unit (CME Corporation, 2026). Industry-wide life-expectancy data for this equipment class shows that the lithium-ion cart battery usually lasts from three to five years of operation, and the cart chassis itself remains serviceable for seven to ten years of service. Any uniform replacement age set within the range of five to six years would be placed right after the battery fails and before the chassis is worn out, meaning that the fleet replaces perfectly good equipment and keeps using equipment whose battery is already dead.

This is not an issue of lack of due diligence; it is a result of using the wrong variable for scheduling replacement. Calendar age serves as an indirect measure of condition of the equipment since a fleet manager knows the age of a cart but does not know whether its battery or scanner are currently working properly. When an institution does not have a continuous condition monitoring system for its device support function, calendar age is the only scheduling variable available, even though it is a poor predictor of the battery and scanner failures that cause clinical disruptions.

The Clinical Consequence

This misalignment has concrete consequences: as shown in a study by Grailey et al. (2023) where behavioral science tools were used to identify BCMA barriers, broken equipment, repair delays, and poor scanner connectivity are among the main obstacles to the BCMA process faced by nurses, with scanners "not reading" and nurses having no means of resolving the issue at the moment (Grailey et al., 2023, Results section). Unreliable BCMA process makes nurses use alternative ways to confirm patient and medication identification, bypassing the barcode scanning procedure. Van der Veen et al. (2018) in a multicenter observational study identified that in 66% of observed cases nurses employed workarounds for medication administrations, and this practice was associated with medication administration errors in a study with the adjusted odds ratio of 3.06 (95% CI: 2.49–3.78), with 78% of those errors being medication omission. Put simply: unreliable BCMA process makes staff find alternative solutions that increase the odds of an error happening.

Regulatory Context for the Clinical Finding

The clinical evidence above is not just a mere operational issue. It is against a specific patient safety standard, and the solution has to be aligned with that standard as well. National Patient Safety Goal 01.01.01 of the Joint Commission requires hospitals to verify a patient's identity by using at least two identifiers, with one of the two not being the room number (Joint Commission, 2024). Barcode scanning of the wristband ID is the operational mechanism through which most academic medical centers, including those comparable to SHC, meet this requirement. A nonfunctional scanner does not just make nurses' work harder; it eliminates, for that particular administration, the required control.

The Service and Repair Model

In this class of equipment, replacement of defective components such as batteries or scanners is normally conducted under a manufacturer or third party warranty and not done through biomedical/IT employees internally. This tends to be a scheduled rather than continuous on-site service call (DTG, n.d.). In cases where this trend applies, the immediate costs incurred with respect to parts and labor for repair of the equipment can be said to be covered in the vendor relationships. The cost faced in this case by the health care institution is that of disruption caused by failure of equipment until the next service call comes by. This is quite a different type of cost than where the institution performs its own repair and pays for it. For this reason, the redesign as suggested in Section VII will be achieved by changing the condition under which service request is made, predicted equipment failure and not failure of the equipment itself. The exact nature of SHC's own service agreement is a key consideration for the internal data request process as noted in Section III, since the extent of the opportunity will depend on the nature of this service agreement.

Why This Matters More at SHC Specifically

It was discussed in Section I that the SHC provides healthcare services beyond its licensed bed capacity with a case mix index of 2.53. The above scenario brings a new dimension to the issue of reliability of equipment, which in a generic situation would be a rather minor matter. An equipment failure in a less acute hospital would have been managed through the slack capacity available within the organization; but for SHC, an equipment failure will happen on units admitting some of the most complicated cases in the area without any slack capacity to act as a cushion for the failure.

Industry Adoption of Condition-Based Models

Condition-based device management is already a proven, not experimental technique for device support in hospitals. The available software for enterprise fleet management targeting hospitals' IT staffs has made it possible to monitor battery conditions, locations of carts, and their utilization patterns and explicitly presents a shift to condition-based, predictive maintenance as its key value proposition (Capsa Healthcare, 2024). Vendors list large, well-funded health systems among their condition-based battery and components management programs adopters, implying that companies like SHC would not be experimenting with something that has not yet been tried out by any other company. Of course, the claims about adopters are coming from vendors themselves with a vested interest in the technology and need to be corroborated independently before using them in an internal business case. But the overall trend is clear – the healthcare industry is moving towards predictive condition-based device management, not away from it.

Adoption along these lines also provides a good benchmarking possibility. Peer institutions that are competing with SHC for grants, physicians, and referrals for more complex cases can be considered likely early adopters of predictive device management. SHC will be able to find out how peer institutions do it via professional connections, vendor recommendations, and the literature on the industry, and thereby benchmark their own initiative against peer practice instead of just believing marketing information from vendors.

III. Financial Analysis and Capital Allocation

The issue posed by this section is a very specific yet crucial one, that is whether changing the criteria for endpoint replacement from age to condition would be economically justified. The answer to the question above, which is explored further in this section, is positive; however, the extent of such an economic benefit is conditional on information that needs to be verified by SHC's internal data procedure.

The Economic Environment

There are three factors which determine whether any capital-adjacent proposal can be advanced in the current period at SHC. First, the capital supply is indeed tight relative to the demand: SHC is currently building out a \$1.6 billion seismic renovation of 300 Pasteur and a \$711 million Redwood City outpatient expansion (S&P Global Ratings, 2025). Second, the organization's 4.74% operating margin is, albeit healthy for an academic medical center, relatively thin to allow much discretionary spending that lacks clear return. Third, SHC's revenue is mainly based on commercial and Medicare fee-for-service and prospective payment (S&P Global Ratings, 2025), which means that avoided operational cost rather than the reimbursement is the main financial benefit of a device reliability initiative, as no payer reimburses the endpoint uptime directly.

Cost Structure of the Current Model

Under the general industry service model described in Section II, the cost of an individual component failure is mostly determined by downtime rather than component cost, as the repairs are typically covered by the vendor warranty. Thus, the cost of the current age-based approach is three compounding inefficiencies rather than just the direct cost of replacement: premature replacement of the chassis, which still has years of useful life left; operation of units with batteries that have exceeded the period of reliable service, generating exactly the failure pattern documented in Section II; and the lack of a leading indicator, meaning that the organization will find out about the issue only once it happens and is reported by the clinician. As none of these three cost categories currently has any quantified measure in this report due to the need for proprietary SHC-specific data, Table 1 in Appendix A outlines the cost categories that would be quantified once the data becomes available, along with the industry benchmarks for each of them.

The Case for a Condition-Based Alternative

The financial case for transitioning to the condition-based replacement lies in the fact that the cost of predicting the component failure and responding to it proactively is lower than the cost of absorbing the consequences of the failure, provided it is possible to act on this prediction using the capacity that is already present in the organization. In case of SHC, if the service program follows the typical scheduled visit approach outlined in Section II, condition-based replacement would not involve new vendor cost, capital expenditure or hiring anyone: it would simply mean that the maintenance schedule would be driven by the condition of the components rather than

their age. That is precisely the kind of proposal well-positioned to receive approval in a capital-tight environment.

Extending the average chassis service life to the upper limit of its 7-10 years range, while doing replacement of the components based on the condition trigger rather than the calendar one, would be expected to result in reduction in the gross annual replacement volume, as the number of prematurely retired chassis will decrease. However, that is somewhat offset by the cost of component replacements done more often and targeted at specific units, but as the replacement cost is generally covered by the warranty, that offset is expected to be relatively small. The scale of this position cannot be estimated confidently in this report, as it is dependent on the actual SHC's fleet size, replacement cadence and unit/component cost, which requires data obtained through the approved data request from Section VII.

The Reimbursement Landscape

It is important to outline clearly what this proposal cannot claim: there is no payer who would reimburse the endpoint reliability. The Hospital-Acquired Condition Reduction Program imposes a payment penalty on hospitals that perform in the bottom quartile of the selected composite of patient safety and healthcare-associated infection measures; medication administration errors related to the equipment failure are not included into that composite. The same applies to the Hospital Value-Based Purchasing Program, which redistributes a portion of withheld base payment depending on the performance; again, the only connection to the equipment reliability is through the patient experience score, at best. The only honest position here is that this proposal should be justified through the avoidance of cost under the existing prospective payment and case-rate reimbursement scheme at SHC, where an additional day of hospitalization due to the preventable adverse event falls entirely on the organization, rather than on a payer. The good thing for the proposal is that in such an environment, the full benefit of the avoided adverse event falls entirely on SHC rather than is shared with or even captured by a payer.

Avoided Harm as an Uncounted Upside

Modeling Approach

Rather than attempt to assert a single dollar figure this report has no basis to substantiate through verified figures, the methodology of this model will be set forth here for the purpose of evaluating the process apart from the specific SHC figures that would be plugged into it. The model will start with the determination of current annual replacement volume and estimation of total fleet size based on that figure and the current average service life, since determining fleet size directly would require performing a fleet reconciliation. From there, it will contrast two potential scenarios, continuation of the status quo age-based policy at current replacement volume versus implementation of the condition-based policy that extends chassis average service life toward the upper end of its reported range while increasing targeted component level replacement volume. The difference between the two annual replacement volume figures, multiplied by verified or benchmark per unit cost, gives a gross annual savings. Against that gross figure, the model will offset any costs for increased component level service volume, any added telemetry platform cost and partial staffing allocation for program management, to yield a net annual position. This structure, gross savings less cost, is precisely what Table 1 in Appendix A is meant to reflect once SHC-specific figures replace the publicly available figures currently populating it.

An Illustrative Worked Example

The model above can become concrete through a fully hypothetical, illustrative example generated using nothing but public benchmark figures, not a single SHC-specific figure is involved. Assume a hypothetical fleet of 1,000 units, selected purely for illustrative purposes as a round number. At 5.5 year average age based replacement cycle, that fleet will require roughly 182 replacements annually. Assuming a condition-based model that extends the average service life of the chassis toward 8 years, still within the 7-to-10 year range reported in the literature (CME Corporation, 2026), the fleet will require approximately 125 replacements annually, a reduction of 57 units. Taking a publicly available list price of \$2,184 for a comparable UL 60601-1 compliant powered mobile workstation (Z&Z Medical, n.d.), the annual gross avoided replacement cost amounts to \$124,500 per year. On top of this, the 57 units no longer being replaced annually will now go through a battery repowering procedure at about half the cost of a new unit, based on vendor reported repower prices (DTG, n.d.), resulting in an increase of \$62,200 in annual component service costs. The result is a net annual positive position of \$62,300 per 1,000 units per year, before including any avoidable patient harm benefit. Table 2 in Appendix A lays out the details of the calculation. This example is included to show that the basic arithmetic works under plausible public figures — but not to provide an estimate of SHC's actual fleet economics, which depend solely on SHC's actual fleet size and contract pricing and will have to be modeled separately once that information is available.

Sensitivity and Risk to the Financial Case

There are three main uncertainties that affect the magnitude of the financial case, though not its direction, and each should be called out rather than brushed under the rug. First, fleet heterogeneity, carts used in high traffic environments like emergency departments and peri-operative areas accumulate mechanical wear at a materially higher rate than carts in low acuity ambulatory areas, so using one uniform service life target for the entire fleet would lead to a substantial understatement of replacement needs for the highest utilizations carts and overstatement of needs for the lowest utilizations ones. A defensible model would use a multi-tiered service life target, and gathering the data needed for setting up those tiers correctly is one of the deliverables of the pilot phase outlined in Section VII. Second, uncertainty of unit and component pricing, without knowing SHC's actual contract terms, any dollar figure in this analysis is necessarily going to be based on publicly available figures, and enterprise procurement would likely have been able to negotiate better prices at SHC's scale, which means that the true economics are even more favorable than a public-benchmark estimate would suggest, not the other way around. Third, uncertainty of service agreement, the entire assumption that condition-triggered service requests can be incorporated into existing capacity at negligible marginal cost hinges on SHC's actual vendor agreement allowing for that kind of a request pattern change, and Section II identifies the agreement as unconfirmed. If it doesn't, then the intervention would need to be renegotiated as part of the vendor's next contract cycle rather than implemented immediately, a very different, yet not an impossible, implementation path.

This analysis intentionally excludes avoidable patient harm from its financial case, and that exclusion itself is a deliberate strategy decision. Bates et al. (1997), in the original cost study of adverse drug events, found that each preventable adverse drug event was associated with additional \$5,857 in cost and 4.6 additional inpatient days. Subsequent analysis performed by Hug et al. (2012) narrowed that down to adjusted figures of \$3,511 and 3.37 additional inpatient days for preventable adverse drug events generally, and \$8,116 and 5.54 days for those deemed

potentially life-threatening. Both figures are expressed in the dollars of their respective publication years and do not take into account the present-day cost levels. Because this analysis excludes these figures from its financial baseline, the recommendation in Section VII does not depend on a contentious harm avoidance calculation to be approved, it is intended to be defensible on economic merits alone, with avoidable patient harm serving merely as an add-on to an already positive case.

IV. Workforce, Leadership, and Collaboration

A proposal that is technically sound but not owned by anyone and not supported by any external partner will never get implemented. This section addresses the human and cross-organizational elements of the proposed shift, the elements that any purely technical analysis inevitably ignores and which make implementation actually possible.

The Workforce Stake

Endpoint reliability is a problem both in terms of technology and retention. In a systematic review and meta-analysis of 66,556 individuals in the literature on clinician burnout associated with electronic health record use, Wu et al. (2024) found that prevalence of burnout in health care professionals linked to electronic health records use was 40.4%, with burnout odds increasing with the time spent on EHR-related activities. Alobayli et al. (2023) independently confirm this link in a systematic review of EHR use and clinician stress and burnout and its influencing factors in hospitals. Workforce economics data make this picture more relevant in terms of retention, the median annual wage of registered nurses is \$93,600 compared to the median annual wage of all occupations which is \$49,500. According to Whalen (2026), among nurses surveyed in 2024 and planning to leave the profession within five years 41% cited stress and burnout as the reason, with retirement being the most common reason. Broken equipment during the medication pass is a small but recurrent source of daily stress for nurses. It does not mean that improving device reliability solves the problem of burnout – the reasons behind burnout go beyond one source of irritation – but it does mean that endpoint reliability belongs inside retention discussions, not outside them, and that the proposal of a solution purely based on technological benefits does not capture its full organizational context.

There is a clear contradiction in the literature above that deserves attention. On the one hand, the findings of Wu et al. (2024) and Alobayli et al. (2023) may be used to make the argument that it is best not to introduce additional technology in the clinical setting to avoid the risk of making people feel even more stressed. However, on the other hand, according to Lee et al. (2025), the cause of failure of technology adoption is "lack of trust, inadequate training, and organisational support" rather than technology itself, which means exactly the opposite, that well-implemented technology that is adopted with the help of the approach described in this paper can reduce the level of burnout. This paper takes the latter perspective, but it makes it explicit rather than ignores the former. The distinction that resolves this contradiction is between the technology that increases the burden of documentation, the subject of EHR-burnout literature, and the technology that gets rid of the recurring source of friction at the bedside, which is the category of the proposed solution. This distinction is an assumption made in this report, not a fact found in literature.

Cross-Functional Governance

Implementing condition-based replacement is likely to require coordination across a range of departments, including device support, nursing leadership, clinical informatics, supply chain, facilities, and risk management, the departments that do not report to each other and that, at SHC, operate from different geographic locations, from Palo Alto to Pleasanton to ambulatory network. Clarifying the governance structure of this organization through which the change will be authorized at SHC is part of the internal data request procedure in Section VII; this report can outline the general structure of the coordination required from the departments involved, but cannot provide information about their reporting lines without confirmation from SHC. One thing can be stated with certainty, approximately 30% of SHC employees are unionized (S&P Global Ratings, 2025), which means that any changes in the medication pass environment, however minor from a technological perspective, should pass through negotiations with the labor representatives and the established SHC nursing shared-governance structure, not just administrative announcements.

Coordination is required in this case not because the departments involved have conflicting priorities but because of structural reasons. the priorities of the departments really contradict each other. Finance and supply chain are structurally oriented towards predictable, deferrable, standardized decisions, fixed replacement age is administratively easy to implement and budget, which is why it is the industry standard (CME Corporation, 2026). Nursing leadership, on the contrary, is structurally oriented towards minimizing bedside disruption whatever the administrative complications, the cost of any failure will fall on their staff and on their patients, not on a budget line. Device support, on its turn, will have to absorb whichever priority will have more organizational leverage. The recommendation in Section VII is essentially a reconciliation of this conflict rather than a victory of one side: it makes the intervention administratively simple and predictable for finance and supply chain, while addressing bedside disruption for nursing leadership, it argues that in this particular case these priorities are not actually contradictory despite the usual conflict, while remaining a potential problem when it comes to data granularity, while finance and supply chain would prefer aggregate reporting, the nursing leadership may need enough unit-level data to advocate for their staff, a problem addressed in the data governance section in Section VI.

Leading the Change: Evidence from the Literature

The literature on health care technology change management provides a consistent finding relevant in this case: adoption of technology is likely to fail in cases of insufficient end-user and leadership involvement regardless of the nature of the underlying technology. According to Lee et al. (2025), in a review of Technology Acceptance Model and Unified Theory of Acceptance and Use of Technology in twenty peer-reviewed articles organizational trust, adequate training, and organizational support were identified as predictors of successful technology adoption in health care organizations, with the usefulness framework being insufficient for clinical settings. It means that success of a condition-based endpoint program depends not on its technological merits but on whether nursing staff can trust that the system helps them reduce bedside disruption and does not increase the monitoring load for them.

Vendor Collaboration as a Leadership Challenge

Perhaps the most interesting thing about the leadership aspect of this proposal is that the key element of it simply cannot be done through internal leadership alone. As noted above, the most effective way to implement the recommended change of service request paradigm is to reach out

to the vendor and agree to a new request process based on device condition rather than failure status, which requires negotiating with a partner not under SHC's full control. Sampson (2024), discussing partnership models in healthcare delivery, notes that technology necessitates treating vendors as strategic partners in healthcare delivery rather than mere service providers: the recommended program is successful only if the relationship with the vendor is treated as such, not as an ordinary service contract between SHC and its vendor.

Diverse and Distributed Teams

It is important to note that the workforce affected by this program is not homogenous, and treating it as such would be counterproductive to the recommended communication method. Clinical workforce of SHC, like of most large urban academic medical centers, is linguistically and culturally diverse, has members from various experience level generations and operates on multiple shift schedules meaning that not everyone is present for a single meeting or an announcement. Change management method that hinges on reaching all people in one centralized session is badly suited to this situation. The proposal that will follow takes multi-channel, repeated, unit-level communication as a necessity, not a nicety. In addition, the supplier chain for this equipment is international: WOW components, batteries, and scanners are manufactured and supplied internationally, and the vendor partnership mentioned above is part of that chain, affected by the same component sourcing, pricing, and availability challenges facing the medical device industry. Therefore, leading this program requires operating successfully in both domains: a domestic diversity of the clinical workforce that needs to be engaged in a person-to-person fashion, and a global supplier chain of equipment vendors and components that needs to be managed at arm's length.

A Communication and Engagement Approach

Consistent with the adoption literature presented above, this report suggests implementing this program with the direct engagement of frontline nurses and unit managers prior to any technology deployment, not after a decision was already made. Lee et al. (2025) specifically mention "a lack of trust, inadequate training, and organisational support" as significant and frequently occurring obstacles to adoption of health care technologies in their review (Lee et al., 2025, Results section). Applied to this case, this finding suggests a particular order of actions: first, briefing nursing shared governance councils and union representatives on the problem described in this report, using general failure pattern described in Section II rather than SHC-specific numbers until those numbers are validated and approved for internal communication; second, selecting unit-level "champions," i.e. nursing staff members who can convince their colleagues of the need for change; and third, piloting the solution in a visible way so that the rest of the organization witnesses the program working on a small scale before being asked to deploy it widely. This sequence has low cost and directly addresses the trust and communication gaps identified as primary drivers of adoption failure, regardless of the quality of the technology involved.

V. Sustainability and Long-Term Asset Strategy

In this section, the proposed strategy is evaluated as a long-term asset strategy rather than an annual budget decision and tested for consistency of financial and environmental arguments. They are consistent.

The extension of the usable service life of clinical endpoint devices is both a cost strategy and an environmental strategy and literature on healthcare increasingly views them as inseparable concepts. Shevchenko et al. (2026), discussing circular economy practices in healthcare, remark that extending the lifespan of complex medical devices is becoming a priority in the sector due to high cost and advanced technological nature of such devices and note that circular design for this kind of equipment requires four qualities: durability, repairability, maintainability and upgradability.

This is directly related to the electronic waste aspect of the proposal. Lithium-ion batteries have different handling, storage-separation, and disposal considerations apart from the generic electronic waste guidelines. A condition-based lifecycle approach whereby batteries are changed before they become unusable as opposed to changing the whole unit including the functioning chassis with a dysfunctional battery will lead to a decrease in the amount of electronic waste created per year of usage and a decrease in the rate at which new units will be manufactured due to replacement needs. These two benefits cannot be quantified without actual fleet and replacement data for SHC, but the direction of the change is clear: a condition-based lifecycle approach results in lower electronic waste created per year of service than an age-based approach where the whole working chassis is discarded together with a dysfunctional battery.

In this light, endpoint lifecycle management is not a one-off process but a recurrent asset strategy that is informed by the condition data collected during the first year — battery condition data, scanner failure rates, chassis lifespan depending on utilization band — and will be a valuable organizational resource in its own right that informs replacement and procurement in the future.

There is also the aspect of community benefit, which should be mentioned in case of a not-for-profit entity. SHC is granted tax exempt status on the basis of providing community benefits, and reduction in the generation of electronic waste and equipment disposal are relevant parts of that broader institutional story. They do not carry the same weight in an environmental, social and governance disclosure that SHC's clinical charity work and mission does, but it is certainly a defensible line item in such disclosure.

VI. Ethical Considerations

The proposal raises ethical issues as much as it solves some, and the thorough ethical analysis should treat them equally rather than present the proposal in the best possible light only.

The Duty Created by Knowledge

The introduction of condition monitoring brings along an ethical responsibility that is not implied by the age-based system. Currently, there is no foreknowledge of the failure of the device, the manager does not know a particular unit will fail soon. The presence of condition telemetry data implies that the organization has evidence that a specific device on a particular unit is deteriorating prior to its failure. Therefore, the further use of the device becomes a choice rather than an oversight that can be claimed after an incident. This is not a reason to abstain from condition monitoring, the other choice of being uninformed is even more unethical, but it means that the telemetry deployment has to be accompanied by the pre-defined response threshold that specifies the level of deterioration required to trigger particular actions in the particular time frame. Otherwise, there will be no reduction in risks but their increase.

Attribution and Just Culture

The next ethical issue concerns the allocation of blame. Any workarounds are obviously an observable behavior of clinicians, whereas the equipment malfunctioning that causes workarounds is often not visible to the person analyzing the incident. This means that there is a real risk of the workaround being blamed on individual nurse noncompliance with the procedures while the direct cause is equipment provided by the organization and planned for the replacement based on its age. According to Grailey et al. (2023), equipment malfunctioning was identified as a structural barrier to BCMA use, not an individual choice of a clinician to not perform a procedure. It means that treating scan-rate variance as the indicator of staff performance, without the investigation of the device condition, means misallocation of the blame from the system to the individual, thus failing the just-culture requirement to investigate possible systemic issues prior to individual blaming.

Distributive Fairness

The third factor that needs to be considered is the distribution of reliability across the population. In case the replacement priorities include unit criticality (which is a reasonable choice, although it cannot be confirmed with the available SHC data), such prioritization can be justified since it directs limited reliability enhancements to those units which are the least tolerant to disruptions. However, it means that reliability will be improved unevenly across the hospital. This is an equity tradeoff that should be transparent for nursing leadership, with clear criteria and rationale for it, rather than an algorithmic process.

Balancing Competing Obligations

Thus, these four factors do not come together in a simple and clear answer. The fact is that knowledge creates an ethical responsibility, which means that the transition to the condition monitoring should be performed, otherwise the organization would continue to remain ignorant. The just-culture concern calls for cautious handling of the data, which would make it possible to prevent new issues caused by mismanaged data while preventing harm related to equipment. The fairness concern highlights the need for transparency of the decision-making process, which requires more effort. Data governance concerns call for moderation in the type of data collected and its level of aggregation. All these concerns do not call for not implementing the program, but for a particular approach in doing so, deploying the capability, but governing it with care, following policies discussed above.

Data Governance

Finally, it is necessary to manage the unit-level reliability data carefully to avoid the situation in which it becomes a tool to monitor the individual clinician behavior. A dashboard that is designed to track the number of times the spare cart is used or scan rate variance is acceptable data for a legitimate purpose. However, the same data disaggregated to the individual level of nurse raises a different privacy and trust concern. This report recommends that the implementation of the system include the explicit definition of the level of data aggregation at which reporting takes place and sticking to this limit.

VII. Recommendations

The recommendations below are arranged in the order from the most important ones to less important ones, they are stated with the degree of specificity provided by the analysis and are conditioned by the data availability.

Alternative Approaches Considered

Three alternatives were considered and ruled out in lieu of the recommendation provided below, and all three merit consideration because it may appear otherwise that the recommendation was the only course of action that was explored. The first of these alternatives involved reducing the age-based replacement schedule to keep pace with battery degradation; this means switching from a five-to-six-year cycle to a three-to-four-year cycle. However, this recommendation would lead to retirement of chassis that have years of life remaining, thus increasing costs and producing more electronic waste without solving the root cause of the measurement issue highlighted above; it deals with the symptom rather than the cause. The second alternative would be to insource the component repairs (i.e., battery and scanner replacement) rather than rely on vendor warranty services for this purpose. This was ruled out due to the necessity of acquiring a new set of capabilities, as well as inventories of spare parts, with a significant likelihood of having to give up existing warranty protection. This makes an easy adjustment a costly effort with a long return on investment horizon. The third alternative would be to do nothing about the current level of failures, as it can be considered a normal business risk. This was ruled out by virtue of the financial and clinical evidence provided in Sections II and III, the current approach involves discarding serviceable equipment prematurely while at the same time accepting an avoidable failure rate, and inaction closes off an improvement avenue that does not entail any investment.

Primary Recommendation

SHC should, in principle, approve the transition of clinical endpoint devices to condition-based replacement, by filling up the available service capacity based on units that have been flagged on the basis of measurements rather than on those that have failed in use. In case the current service arrangement for SHC involves the standard warranty and scheduled visit process described in Section II, this will be a relatively small change in operations – it will involve no additional spending with vendors, no insourcing of repairs, and no additional nursing procedures. Approval in principle will allow beginning planning and vendor involvement without having to wait for the data that currently does not exist in validated form.

Supporting Recommendations

- Shift the basis of replacement authorizations from calendar age to measured condition of components, such that the chassis will be retired closer to the end of their service life rather than according to a standard age cutoff point, and the replacement target will be tied to unit utilization rather than applying uniformly to the entire fleet.
- Establish the accuracy of device records and configurations to the necessary extent in order to attribute condition measures to a particular unit and its location; do so first, because everything else in this recommendation depends on it. This is not just a new project cost, the requirement for keeping records of hardware and media and of persons responsible for their movements under HIPAA Security Rule Accountability Implementation Spec. (45 C.F.R. § 164.310(d), 2017).
- Involve the equipment service vendor in rethinking the service request initiation procedure and look for relevant provisions in the current service agreement.
- Develop a written response-threshold policy, identifying at what measurement state what action is to be taken, as an approved deliverable that must exist prior to any activation of

telemetry on condition monitoring, in accordance with the ethical considerations of Section VI.

- Define the level of aggregation of the reliability reporting (by unit, not individual clinician) prior to design of the dashboard.

The Internal Data Request

While this document was designed to stand alone without further data, its financial accuracy would benefit from SHC-specific confirmation of the following: current size of fleet; current weekly/monthly replacement quantity; details of the vendor service agreement, including how repairs are handled (vendor versus internal, and on what schedule); per unit and per component costs; and whether any current tracking exists of weekly failures in number. Each of these is subject to the confidentiality clause of this project and should be obtained via internal processes rather than presumption. Table 1 in Appendix A offers a placeholder financial model based on public benchmark pricing, deliberately constructed such that each public estimate is easily substituted with a confirmed SHC number.

Implementation Sequence

Phased implementation is suggested over immediate full-scale rollout, in accordance with the innovation and strategic practice principle that innovative and strategic approaches to business practices should challenge the status quo while acknowledging change management implications. Phase one will focus on configuration data remediation and vendor engagement, which requires neither additional telemetry nor any investment of capital. Phase two will pilot the use of condition monitoring in a select segment of the fleet, preferably including some units of high acuity as well as lower-acuity units, to gain SHC-specific data on failure patterns and test out the response threshold policy in a controlled environment prior to enterprise rollout. Phase three, contingent upon phase two, will roll out the program enterprise wide with tiered service life goals informed by the SHC data collected in the pilot. Such sequencing provides the opportunity to validate the model's assumptions against SHC data prior to large-scale investment, which addresses the single most important limitation of a proposal based largely on public rather than internal data.

Risk Management

The following four risks require proactive management rather than mere recognition. The risk of highest probability is that asset and configuration data are insufficient to permit accurate condition attribution; the mitigation is addressed above, with the sequencing of phase one before telemetry investment. The risk of highest consequence is implementation of condition monitoring without the response threshold policy identified in Section VI as an ethical precondition; the mitigation is for the policy to become a hard gate for telemetry activation, not a parallel effort that may lag behind the technical effort. A third risk is the vendor refusing to offer a condition-based service model in terms of their agreement; the mitigation is vendor engagement as a separate phase one workstream, which will permit discovery of a negative result early and cheaply, rather than later when other investments have been made. A fourth risk is that the lack of a dedicated headcount results in decay of the program over time due to day-to-day pressure; the mitigation is to name a specific program manager accountable to both nursing and device support, and to measure success quarterly against the criteria below.

Stakeholder Impact Summary

It involves a larger group of stakeholders than device support alone, and calling them all out explicitly lowers the risk that someone will get forgotten as part of the implementation process. Nurses face less breakdown of bedside devices and, should they follow the communication strategy outlined in Section IV, are involved as partners in the change process rather than subjects of it. Device support and biomedical engineering staff have their task load shift from being primarily reactive dispatch to having regularly scheduled, predictable maintenance needs. This is a change in nature rather than volume of the work that they do. Clinical informatics now has another durable data asset in the form of condition telemetry, and with that another responsibility to govern that data appropriately. Finance and supply chain staff see the cost of replacements shift from a calendar-based lumpy distribution to a condition-based one that distributes more evenly. Risk management and legal departments receive a new artifact, the response-threshold policy, that previously did not need to exist, with the corresponding responsibility to ensure that it is enforced. Finally, patients are supposed to be the beneficiaries of all of this, although they happen to be the stakeholders in the least position to advocate on behalf of the change themselves, which is precisely why the interest of patients has been treated in this report as a constant and not just one of the voices competing in the recommendation process.

Success Measures

The success of the recommended changes should be evaluated according to the following few clear indicators: the reduction in the amount of discovered device failures as compared to pre-implementation baseline, once the baseline is established on SHC's own terms; the proportion of component replacements caused by detected condition versus failure or age of the device; and, depending on the agreement of SHC nursing leadership on this point, any changes in the BCMA workaround frequency on the unit level. None of the above indicators require disclosure of any additional information beyond what SHC already has approved for the use in this analysis, and all can be reported on an aggregated level consistent with the data governance policy outlined in Section VI.

Limitations of This Analysis

There are three important limitations of this report that need to be considered when implementing its recommendations: first, and most importantly, this analysis relies heavily on publicly available industry benchmarks and peer-reviewed literature because of the terms under which SHC has allowed this project, as the inclusion of any confidential information, proprietary operational data, and non-public metrics requires further approval. All of the directions in this report, that age-based replacement is mismatched to component degradation, that vendor-warranted scheduled service provides an easy way to perform predictive intervention, that financial case does not rely on a harmful side-effect to be positive, are based on well-established patterns in industry and clinical literature. The magnitude of the opportunity at SHC specifically cannot be quantified accurately until the internal data request described in Section VII is obtained and approved. Second, this report assumes, based on common industry practice, that the device service process at SHC follows the vendor-warranted, scheduled visit model described in Section II. In case if this assumption turns out to be incorrect. For example, if the repairs are performed internally, or if the maintenance process is continuous rather than scheduled, the particular mechanism of the recommended intervention would need to be rethought, but the problem itself remains unchanged. Third, this report does not include any primary interviews

with SHC nurses, device support technicians, or leadership and relies instead on peer-reviewed literature about clinician experience with barcode medication administration and EHR burnout. The primary data collection at SHC specifically, conducted via the internal process, would substantially improve both the diagnostic and recommendation portions of any future phases of this analysis.

Conclusion

Clinical endpoint devices managed by Stanford Health Care are operated according to a calendar-age replacement schedule that is structurally mismatched to the real rate of degradation of the components involved. It is a pattern well documented in general industry and clinical literature and further supported by SHC's internal data once the internal data request outlined in Section VII is completed. As a result, SHC uses premature replacement of functioning devices, and leaves degrading batteries and barcode scanners in the care of patients until they fail. There is a well-documented effect on the reliability of BCMA, and therefore, on patient safety. However, a condition-based replacement schedule that uses existing already-funded service relationship, rather than building new capacity from scratch, can turn this situation around and provide SHC both with capital efficiency and clinical benefits, once SHC confirms the exact figures that this report could not quantify. Therefore, the recommendation is not to delay implementing the proposed changes until all necessary information is collected, but to allow this program in principle immediately, subject it to confirmation via internal data request and pilot program, and implement response threshold and data governance policies identified in this report as ethical prerequisites without exception. It allows SHC to transform a reactive maintenance operation into an asset management function without breaking its capital constraints and confidentiality terms.

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Appendix A: Supporting Exhibits

Table 1

Cost Categories of the Present Age-Based Replacement Framework

Cost category	SHC data needed	Public benchmark available in the interim
Premature chassis retirement	Fleet size; current replacement age	Chassis service life 7–10 years (CME Corporation, 2026)
Reactive failure downtime	Weekly failure count; service turnaround time	Vendor-warranted, scheduled-visit service is common in this equipment category (DTG, n.d.)
Component replacement cost	Per-unit and per-component contract pricing	Public list pricing for comparable equipment
Avoided harm (excluded from base case)	Not required; presented as literature context only	\$3,511–\$8,116 per preventable adverse drug event (Hug et al., 2012)

Note. Table used to structure the cost assessment outlined in Section III. Public benchmarks provided as reference; SHC-specific information must be approved by the data request process prior to inclusion in any capital investment proposal.

Table 2

Worked Example of Illustration:

Line item	Basis and assumption	Annual amount
Illustrative fleet size	Hypothetical, round number for illustration only	1,000 units
Current replacements (5.5-yr cycle)	$1,000 \div 5.5$	~182 units/yr
Proposed replacements (8-yr cycle)	$1,000 \div 8$; within 7–10 yr chassis range (CME Corporation, 2026)	~125 units/yr
Gross avoided replacement	57 fewer units $\times$ \$2,184 (Z&Z Medical, n.d.)	+\$124,500
Component repower cost	57 units $\times$ \$1,092 (half of new-unit price, per DTG, n.d.)	–\$62,200
Illustrative net position	Before any avoided-harm benefit	+\$62,300

Note. The table represents a completely fictional example, which uses a round fleet number and publically available pricing information only. It is meant to demonstrate how the modeling approach described in Section III can be applied and does not constitute estimation of any actual SHC figures. To build such a model based on SHC's real figures, additional research will be required.

Table 3*Phases of Implementation and Measures of Success*

Phase	Focus	Principal actions	Success measure
1	Foundation	Configuration data remediation; formal vendor engagement; response-threshold policy drafted	Data accuracy threshold met; vendor agreement reviewed
2	Pilot	Condition monitoring deployed on a limited, mixed-acuity segment of the fleet	SHC-specific failure and service data generated; threshold policy tested
3	Scale	Enterprise rollout with tiered, utilization-based service-life targets	Reduction in bedside-discovered failures against confirmed baseline

Note. See Section VII. The timeframe information has been deliberately left out pending the validation of the SHC's current replacement schedule.